Can semantic feedback improve RTL verification?

Use verified E-graph groups to make
LLM verification feedback more actionable.

01
Coverage efficiency

02
Monitor correctness

03
Fault detection

Coverage ≠ correctness. Semantic classes ≠ RTL coverage points.

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One tool loop, two verification tasks

Implemented prototype: Python orchestration + Rust egg + standard EDA tools

Frozen inputs

Specification + RTL
Fixed task contract

GPT generation

Stimulus JSON or
passive SV monitor

Compile + check

Icarus / Verilator
Yosys + SMT diagnostics

Tool feedback

Actual tool counters
Independent grading

C adds: typed guards → restricted E-graph groups → Z3 equivalence checks

TRACK S Generate stimuli; measure real RTL coverage.

TEST2 Fix stimuli; generate monitors; detect hidden faults.

Claude reference ≠ GPT candidate. Two-state guard proof ≠ sequential / four-state proof.

B and C differ in feedback, not the test bench

Latest test2 cohort · fixed-stimulus passive monitor generation

B **Raw feedback**

Raw coverage + shared development feedback

C **Semantic feedback**

Everything in B + verified E-graph guard groups

SAME DUT · Claude checker · stimuli · gpt-6.1-sol / low

3 sessions / arm × 6 rounds · 64k input / 8192 output caps

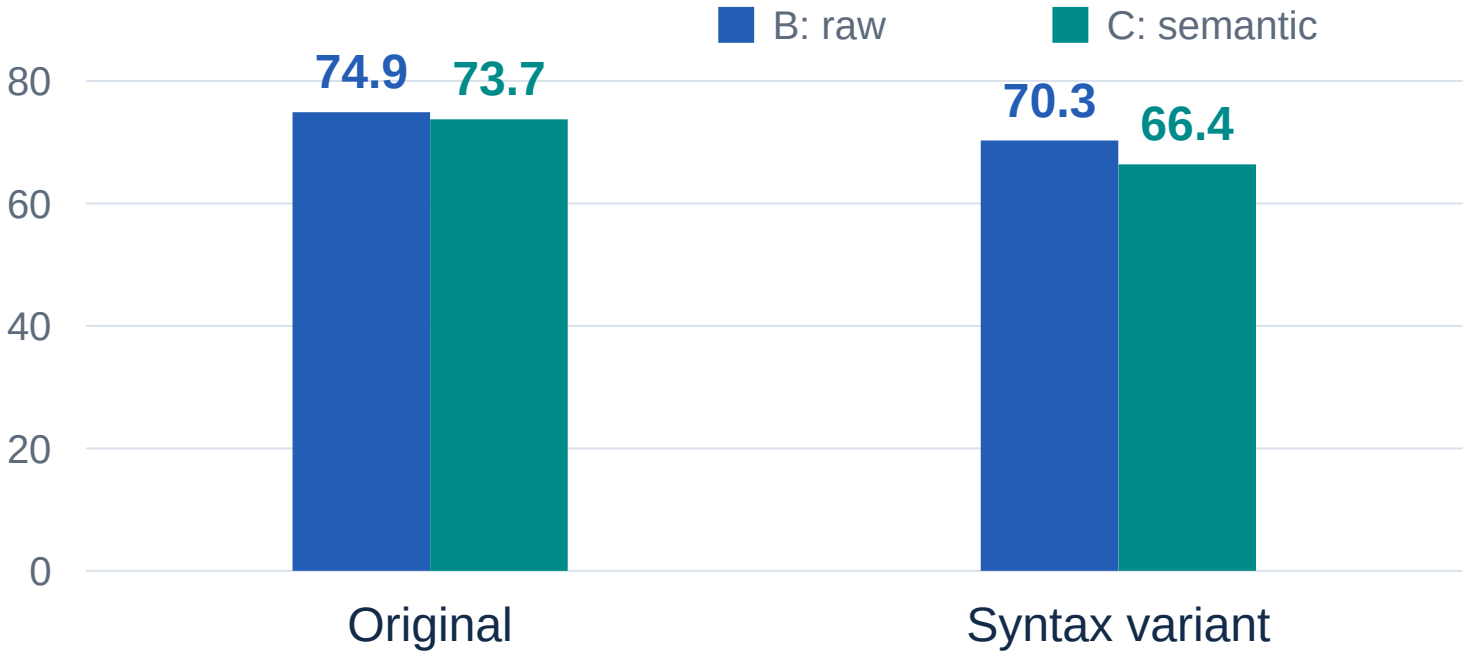
Generate 36 candidates —— **Freeze every round** —— **Hidden grading**

No hidden faults, evaluation traces or reference code in prompts.
Equal caps ≠ equal actual tokens; no token-matched placebo in this cohort.

Queue pilot: a small, mixed efficiency signal

gpt-6-sol · 1 queue IP, 2 RTL forms · 3 sessions / arm / form · 4 rounds

Mean API tokens to reach 41/41 line/case points (thousands)



2 / 6

C earlier by 1 round
4 comparisons tied

Same final coverage in all 12 runs

Final: line/case 41/41 · branch 14/14 · toggle 344/408

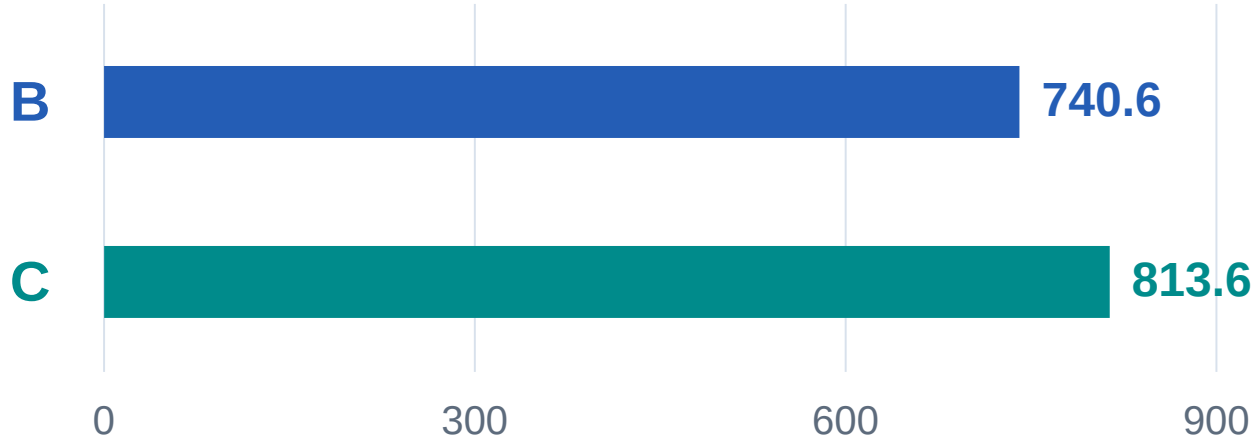
Exploratory, not a causal result: selected guard members were already hit after round 1.

Source: queue_bc_gpt6sol_20261026/target_costs.csv; index-matched comparisons, not paired RNG trials

test2: equal final detection, higher C token cost

Latest trace_done cohort · 3 sessions / arm × 6 rounds · all 6 final monitors

Total API tokens, all 18 rounds / arm (thousands)



11 / 11

observable fault types

C: +9.86% tokens

No extra final fixed-trace kills.
N10 stays in 12 raw types.

Formal: 75 PASS · 104 FAIL · 361 TIMEOUT / 540 checks

FAIL modes: 94 BMC · 7 prove · 3 cover

94 BMC → 20 physical alarms + 74 unreproduced / unresolved (all C)

Secondary prefix-type union: B 3 / C 7; no same-sample oracle confirmation or 128-cycle proof.

Next: remove the ceiling, then test attribution

A working pipeline does not yet establish a general E-graph advantage.

01 **Unsaturated temporal obligations**

Use held-out cases that separate safety from finite completion.

02 **Token-matched placebo**

Separate semantic value from extra feedback and token spend.

03 **Qualify formal-to-physical mapping**

Resolve four-state representation limits; retain unknowns.

Proposed for Nov 9 P2; R0/R1/R2 recovery comparisons remain planned.

Feedback: prioritize temporal completion, or a semantics-only ablation?